

UEEOS Master Specification v3.0 - Universal Evolution & Discovery Ecosystem

UEEOS MASTER SPECIFICATION v3.0

VISION

A universal evolution ecosystem capable of understanding, discovering, simulating, governing, regenerating, and evolving systems from genes and ecosystems to enterprises, economies, societies, and civilizations.

1. UNIVERSAL REALITY FABRIC

- Physical Systems
- Biological Systems
- Ecological Systems
- Human Systems
- Digital Systems
- Economic Systems
- Civilization Systems
- Planetary Systems

2. UNIVERSAL EVOLUTION STACK

Reality → Observation → Knowledge → Intelligence → Wisdom → Purpose → Action → Learning → Adaptation → Evolution → Regeneration → Discovery

3. UNIVERSAL SCIENTIFIC METHOD ENGINE

Observe

Hypothesize

Experiment

Measure

Validate

Learn

Evolve

4. AUTONOMOUS DISCOVERY ENGINE

- Hypothesis Generation
- Experiment Design
- Simulation Testing
- Evidence Validation
- Theory Formation

5. PHYSICS & EARTH SYSTEMS ENGINE

- Hydrology
- Energy Systems

- Climate Systems
- Atmospheric Systems
- Geospatial Systems

6. MOLECULAR BIOLOGY ENGINE

- Genomics
- Proteomics
- CRISPR Models
- Population Genetics
- Evolutionary Biology

7. PLANETARY BOUNDARIES ENGINE

- Climate Boundaries
- Nitrogen Cycle
- Phosphorus Cycle
- Water Systems
- Biodiversity Limits

8. REGENERATIVE INTELLIGENCE ENGINE

- Carbon Sequestration
- Circular Systems
- Ecosystem Restoration
- Sustainable Development

9. UNIVERSAL KNOWLEDGE FABRIC

- Ontology Engine
- Knowledge Graph
- Semantic Layer
- Vector Memory
- Digital Thread

10. UNIVERSAL DIGITAL TWIN FABRIC

- Human Twin
- Farm Twin
- Enterprise Twin
- Industry Twin
- Nation Twin
- Civilization Twin
- Planetary Twin

11. UNIVERSAL AGENT SOCIETY

- Scientist Agents
- Agriculture Agents
- Enterprise Agents

- Governance Agents
- Simulation Agents
- Evolution Agents
- Meta Agents

12. UNIVERSAL SIMULATION LAB

- Enterprise Futures
- Economic Futures
- Climate Futures
- Agricultural Futures
- Civilization Futures

13. COLLECTIVE INTELLIGENCE NETWORK

Humans + AI + Organizations + Institutions + Research Networks

14. UNIVERSAL GOVERNANCE & STEWARDSHIP

Policy, Ethics, Risk, Compliance, Sustainability, Regeneration

15. MATURITY MODEL

L1 Reactive

L2 Managed

L3 Connected

L4 Intelligent

L5 Autonomous

L6 Evolutionary

L7 Regenerative

L8 Discovery-Driven

L9 Meta-Evolutionary

16. NORTH STAR

Enable every system to continuously learn, discover, adapt, regenerate,
and evolve toward greater resilience, sustainability, intelligence, and flourishing.